

Producer Theory

Lecture 10: The Geometry of Costs — Short Run vs Long Run

Paulo Fagandini

2026

Welcome Back! 🖐️

We've crossed the aisle:



Lectures 5–9: **Consumer Theory** (demand side)



Lectures 10–15: **Producer Theory** (supply side)



Today's Goal: Understand the **cost structure** of firms

How do costs behave as production changes?

From Consumer to Producer

The Other Side of the Market

Consumer (Lectures 5–9):

- Maximizes **utility** 
- Subject to **budget constraint**
- Generates **demand curves**
- Tools: indifference curves, MRS, budget line

...

 **Key parallel:** Just as consumers face trade-offs with limited income, producers face trade-offs with limited resources and technology.

Producer (Lectures 10–15):

- Maximizes **profit** 
- Subject to **cost structure**
- Generates **supply curves**
- Tools: cost curves, marginal analysis

The Producer's Problem

THE PRODUCER'S GOAL

A firm wants to **maximize profit**. To do so, it must understand its costs at every level of output.

$$\pi = \text{Total Revenue} - \text{Total Cost} = TR - TC$$

Before we can find the profit-maximizing quantity (next lecture), we need to master costs.

Today we answer:

- **1** What types of costs does a firm face?
- **2** How do costs change as output increases?
- **3** What is the difference between short run and long run?

Short Run vs Long Run

Defining the Time Horizons

SHORT RUN VS LONG RUN

Short run: A period in which **at least one** factor of production is **fixed** (cannot be changed). Typically, capital (machines, buildings) is fixed.

Long run: A period long enough that **all** factors of production are **variable**. The firm can adjust everything: equipment, factory size, workforce.



Short Run — Tourism example:

A hotel has 100 rooms. It **cannot build more rooms** this season. It can only hire/fire staff (variable input).



Long Run — Tourism example:

The hotel chain can build a **new wing**, open in a **new city**, or **exit** the market entirely.

Inputs: Fixed vs Variable

Fixed Inputs

Variable Inputs

Definition

Cannot change in the short run

Can be adjusted at any time

Examples

Building, machines, lease contracts

Workers, raw materials, energy

Tourism

Hotel building, aircraft, kitchen equipment

Staff, food supplies, fuel

Cost behavior

Same cost regardless of output

Cost changes with output

...

THE KEY DISTINCTION

In the **short run**, the firm has both fixed and variable inputs. In the **long run**, **all** inputs are variable — there are no fixed costs.

The Law of Diminishing Returns

LAW OF DIMINISHING RETURNS

When we keep adding more of a **variable input** (e.g., workers) to a **fixed input** (e.g., a kitchen), eventually each additional worker adds **less and less** to total output.

Tourism example: A restaurant kitchen with 5 stoves 🔍

Extra chefs hired	Extra meals/hour produced	Phase
1st chef	+10 meals	Initial Setup
2nd chef	+25 meals	Increasing returns (Specialization)
3rd chef	+18 meals	Diminishing returns (Space starts to limit)
4th chef	+10 meals	Further diminishing
5th chef	+3 meals	Crowding (Stoves are fully occupied!)



Why? The fixed input (kitchen space) gets **congested**. Chefs bump into each other, wait for stoves, compete for prep space.

Types of Costs in the Short Run

The Cost Family

All short-run costs derive from three building blocks:

Fixed Cost (FC)

Costs that **do not change** with output.

Must be paid even if $Q = 0$.

Hotel: mortgage, insurance, property tax

...

 **Important:** FC is a **sunk cost** in the short run — it cannot be recovered. Remember Lecture 3!

Variable Cost (VC)

Costs that **change** with the level of output.

Zero when $Q = 0$.

Hotel: staff wages, laundry, food, energy

Total Cost (TC)

The sum of all costs.

$$TC = FC + VC$$

Hotel: everything combined

A Worked Example: Algarve Beach Bar

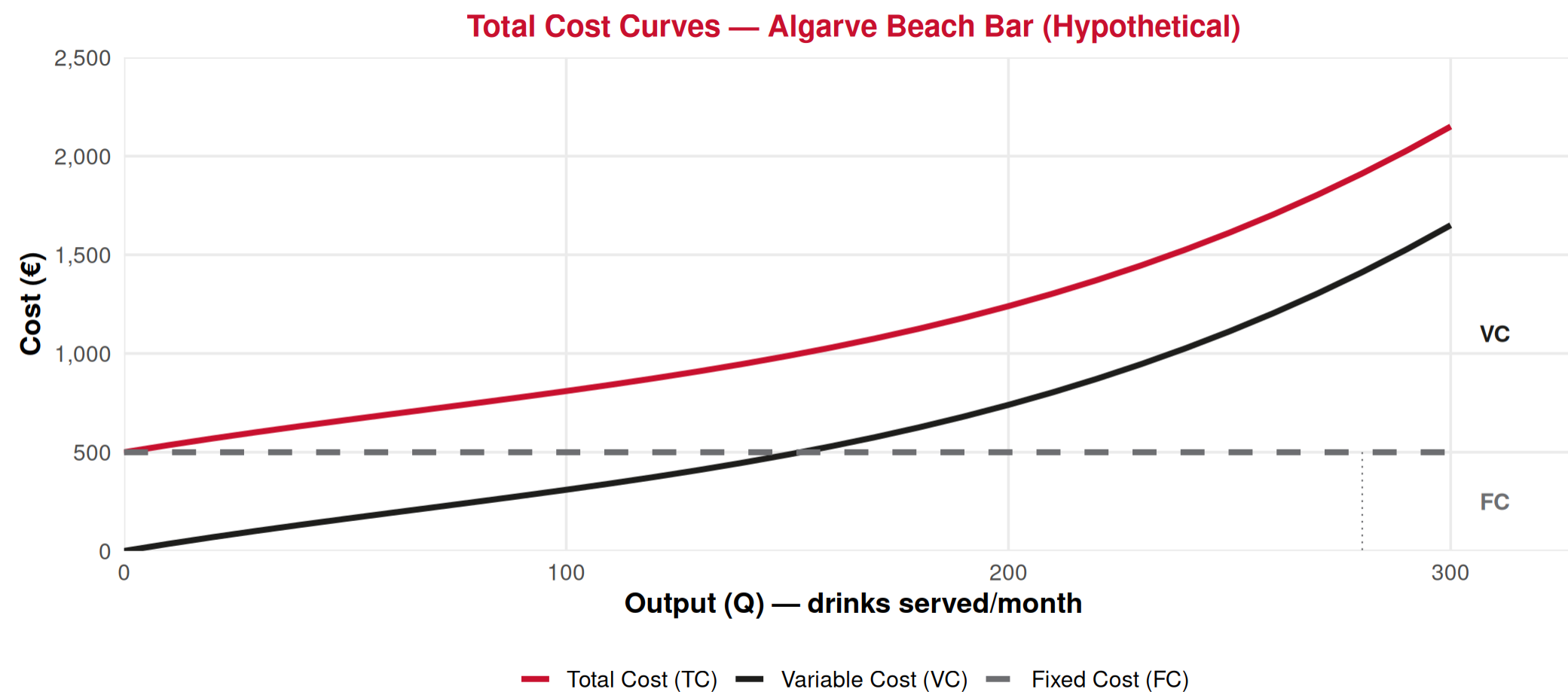
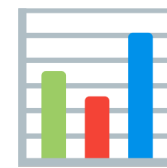
Imagine a small beach bar in Albufeira. The owner pays **€500/month** in rent (fixed), and hires staff and buys supplies (variable).

Output (Q)	Fixed Cost (FC)	Variable Cost (VC)	Total Cost (TC = FC + VC)
0	€500	€0	€500
50	€500	€200	€700
100	€500	€350	€850
150	€500	€550	€1,050
200	€500	€800	€1,300
250	€500	€1,150	€1,650
300	€500	€1,650	€2,150

Output = drinks served per month (hypothetical illustrative example)

👉 Notice: **FC stays at €500** always. **VC increases** — and it increases **faster and faster** (diminishing returns to the staff in the small bar!).

Plotting the Total Cost Curves



👉 **TC and VC have the same shape** — TC is just VC shifted **up by FC**. The vertical gap between TC and VC is always exactly **€500** (the fixed cost).

Average and Marginal Costs

Why Averages Matter 🤔

Total costs tell us the **whole bill**, but firms need to know the **cost per unit** to set prices and make decisions.

AVERAGE COSTS

$$AFC = \frac{FC}{Q} \quad AVC = \frac{VC}{Q} \quad ATC = \frac{TC}{Q} = AFC + AVC$$

Average Fixed Cost (AFC):

- FC spread over more units
- **Always falls** as Q increases
- 🏠 A hotel's mortgage cost per guest falls as occupancy rises

Average Variable Cost (AVC):

- Variable cost per unit
- Falls first, then **rises** (due to diminishing returns)
- 🍴 Food cost per meal — efficient at moderate volume, rising when kitchen is overloaded

Marginal Cost: The Most Important Curve

MARGINAL COST (MC)

The **additional cost** of producing **one more unit** of output.

$$MC = \frac{\Delta TC}{\Delta Q} = \frac{\text{Change in Total Cost}}{\text{Change in Output}}$$

Equivalently: $MC = \frac{\Delta VC}{\Delta Q}$ (since FC doesn't change!)

Intuition: If producing 100 drinks costs €850 total and producing 101 drinks costs €856, then $MC = €6$.

Why is MC so important?

- It tells the firm whether producing one more unit **adds to or subtracts from** profit
- Rule of thumb: **produce more if the price you receive > MC**
- MC is the **foundation of the supply curve** (Lectures 12–14)

Computing Averages and MC: Beach Bar

Q	FC	VC	TC	AFC = FC/Q	AVC = VC/Q	ATC = TC/Q	MC = $\Delta TC / \Delta Q$
0	500	0	500	—	—	—	—
50	500	200	700	10.00	4.00	14.00	4.00
100	500	350	850	5.00	3.50	8.50	3.00
150	500	550	1,050	3.33	3.67	7.00	4.00
200	500	800	1,300	2.50	4.00	6.50	5.00
250	500	1,150	1,650	2.00	4.60	6.60	7.00
300	500	1,650	2,150	1.67	5.50	7.17	10.00

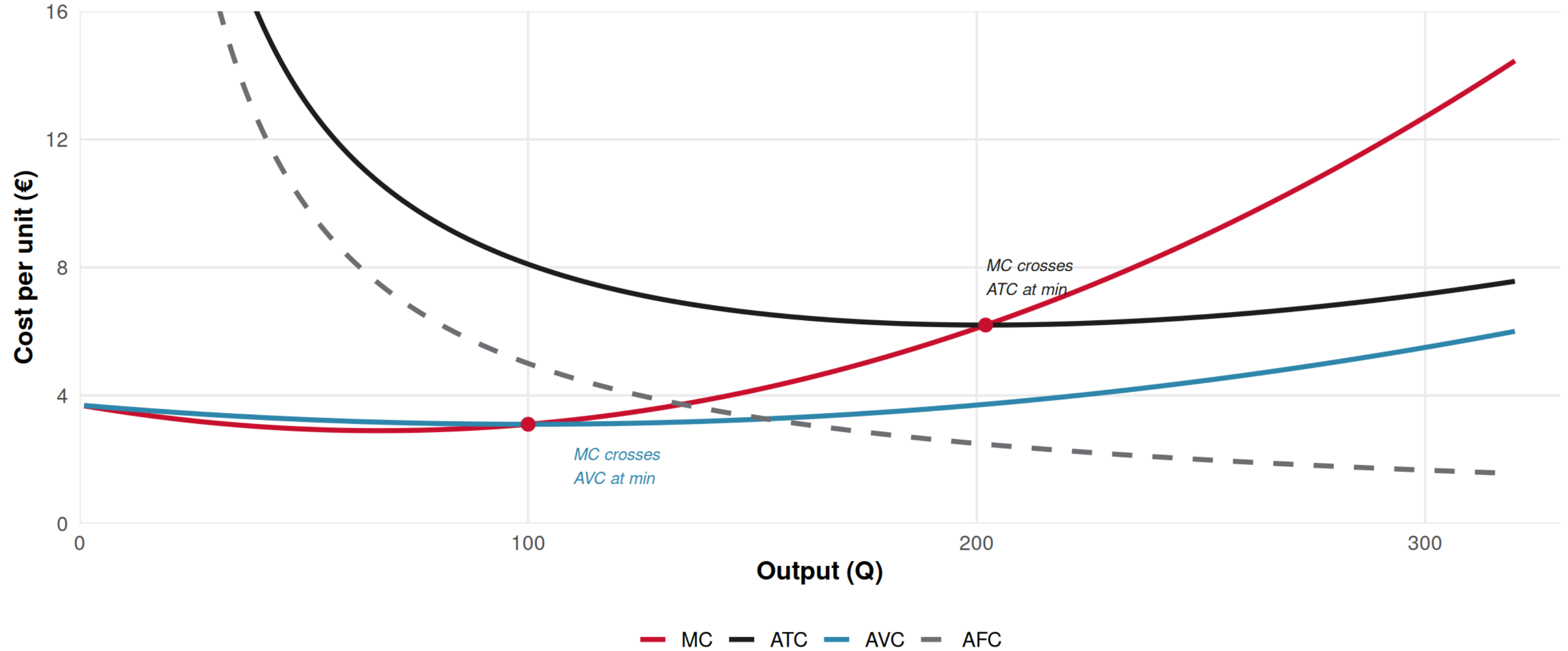
Hypothetical illustrative example

Key observations:

- **AFC** always falls (spreading the fixed cost)
- **AVC** falls then rises (U-shaped) — minimum at Q = 100 (AVC = €3.50)
- **ATC** falls then rises (U-shaped) — minimum at Q = 200 (ATC = €6.50)
- **MC** falls then rises — and **crosses AVC and ATC at their minimums!**

The Geometry of Costs

Average and Marginal Cost Curves (Hypothetical Beach Bar)



Why Does MC Cross AVC and ATC at Their Minimums?



Think of it like your **grade point average** 🎓

The “exam grade” analogy:

- Your **average grade** = ATC (or AVC)
- Your **next exam grade** = MC

If your next exam grade is **below** your average → your average **falls** ↓

If your next exam grade is **above** your average → your average **rises** ↑

If your next exam grade **equals** your average → average **stays the same** (minimum!)

Applied to costs:

- When $MC < AVC \rightarrow$ AVC is **falling** ↓
- When $MC > AVC \rightarrow$ AVC is **rising** ↑
- When $MC = AVC \rightarrow$ AVC is at its **minimum**

Same logic applies to ATC!

👉 **MC always crosses average curves at their lowest point.**

This is a **mathematical fact**, not an economic assumption.

The Shapes You Must Know

Total Cost Curves:

- **FC**: Horizontal line (constant)
- **VC**: Starts at origin, S-shaped (first rises slowly, then steeply)
- **TC**: VC shifted up by FC (same S-shape)

The S-shape comes from diminishing returns — costs eventually accelerate.

...

 **Don't confuse**: MC can be **below** ATC while ATC is still **falling**. ATC rises only after MC **crosses above** it.

Per-Unit Cost Curves:

- **AFC**: Always declining (hyperbola)
 - **AVC**: U-shaped (falls then rises)
 - **ATC**: U-shaped, above AVC (the gap is AFC)
 - **MC**: U-shaped, crosses AVC and ATC at their minimums
-  As Q grows, ATC gets closer to AVC because AFC shrinks.

Tourism Application

Costs in the Hotel Industry 🏨

Fixed costs (don't change with occupancy):

- 🏠 Mortgage / building lease
- 🛡️ Insurance
- 📄 Property tax
- 💻 Booking system software license
- 👤 Core management salaries
- 🛠️ Scheduled maintenance

These costs explain why hotels hate **empty rooms** — FC is already committed!

...

💡 The difference between FC and VC explains why **low-season pricing** exists: covering variable costs and **some** fixed costs is better than covering **none**.

Variable costs (change with occupancy):

- 🧼 Cleaning and laundry per room
- 🔌 Energy and water per guest
- 🥐 Breakfast/meal supplies
- 🛏️ Linen and amenity replacement
- 👥 Part-time/seasonal staff

That's why hotels offer last-minute discounts — as long as $P > MC$ of hosting one more guest, it's worth it!

Spreading Fixed Costs: Why Airlines Overbook

The airline cost structure (illustrative):

An Algarve charter flight has roughly:

-  **High fixed costs:** aircraft lease, crew salaries, airport fees, fuel (mostly fixed per flight)
-  **Low variable costs per passenger:** meals, baggage handling, marginal fuel

...

SPREADING THE FIXED COST

If a flight costs €20,000 to operate (mostly fixed) and has 200 seats:

- **100 passengers:** AFC = €200/passenger
- **150 passengers:** AFC = €133/passenger
- **200 passengers:** AFC = €100/passenger

Every additional passenger **dramatically lowers** the cost per passenger!

 This is why airlines practice **overbooking** and **dynamic pricing** — filling seats spreads the enormous fixed cost.

Long Run Costs

In the Long Run, Everything is Variable

In the **long run**, the firm can adjust **all** inputs — including building size, number of machines, and technology.

LONG-RUN AVERAGE COST (LRAC)

The lowest possible average cost for each level of output, when the firm is free to choose the **optimal scale** (size) of operations.

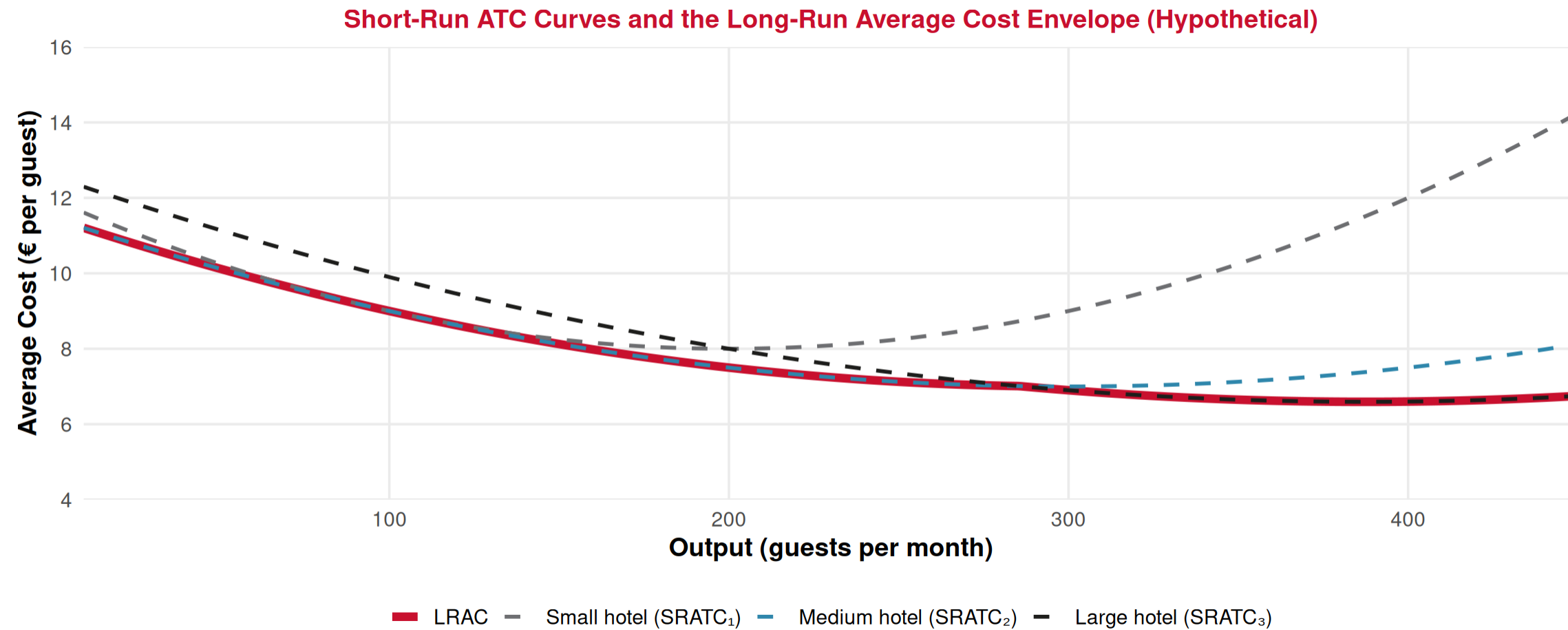
Think of it this way:

- **Short run**: “Given our current hotel (100 rooms), what’s our cost?”
- **Long run**: “What **size of hotel** minimizes our cost per guest?”

The LRAC curve is formed by choosing the **best short-run plant size** for each output level.

👉 The LRAC is the “**envelope**” of all possible short-run ATC curves.

The Envelope: LRAC as the Best of All Short Runs



Each dashed curve = one plant size (short run). The **bold red LRAC** = lowest cost achievable at each Q.

👉 In the long run, the firm **chooses the plant size** that minimizes cost for its desired output.

Economies and Diseconomies of Scale

The shape of the LRAC curve reveals three zones:

Economies of Scale

LRAC is **falling**

Bigger → cheaper per unit

Tourism: Large hotel chains get bulk discounts, shared booking platforms, brand recognition

“Getting bigger makes us more efficient”

↔ Constant Returns

LRAC is **flat**

Doubling inputs doubles output (cost per unit stays the same)

Tourism: Mid-sized operators replicating a proven format across locations

“We’re at the right size”

Diseconomies of Scale

LRAC is **rising**

Bigger → more expensive per unit

Tourism: Huge resorts with bureaucracy, coordination problems, slow decision-making

“We’ve grown too big”

...

 **Minimum Efficient Scale (MES):** the smallest output where LRAC reaches its minimum. Firms should aim to operate **at or beyond** this point.

Economies of Scale in Tourism: Real-World Logic

Why do big hotel chains (Marriott, Accor, Hilton) have cost advantages?

Sources of economies of scale :

-  **Bulk purchasing**: negotiate lower prices with suppliers (linens, food, amenities)
 -  **Technology**: one central booking system serves thousands of properties
 -  **Marketing**: global advertising cost spread over many hotels
 -  **Training**: standardized programs cheaper per employee
 -  **Financing**: large firms borrow at lower interest rates
-  **Bureaucracy**: decisions take longer in large organizations
 -  **Communication**: harder to coordinate across 8,000+ properties
 -  **Loss of local identity**: cookie-cutter hotels may lose appeal
 -  **Monitoring costs**: ensuring quality across locations
-  This is why **boutique hotels** can survive alongside chains — they operate efficiently at a smaller scale with a different value proposition.

Putting It All Together

Summary of Cost Curves

Concept	Formula	Shape	Key Fact
Fixed Cost (FC)	Constant	Horizontal line	Doesn't change with output
Variable Cost (VC)	Changes with Q	S-shaped (rises, accelerates)	Due to diminishing returns
Total Cost (TC)	$FC + VC$	VC shifted up by FC	
Avg Fixed Cost (AFC)	FC / Q	Always declining	Spreading the overhead
Avg Variable Cost (AVC)	VC / Q	U-shaped	Min where MC crosses it
Avg Total Cost (ATC)	$TC / Q = AFC + AVC$	U-shaped (above AVC)	Gap to AVC = AFC (shrinks)
Marginal Cost (MC)	$\Delta TC / \Delta Q$	U-shaped	Crosses AVC & ATC at their mins
LRAC	Envelope of SRATCs	U-shaped	Economies → constant → diseconomies

Summary

Today's Key Takeaways:

1. **Short run**: at least one input is fixed; **Long run**: all inputs are variable
2. **Total costs**: $TC = FC + VC$ — fixed costs don't depend on output
3. **Average costs**: $ATC = AFC + AVC$ — both per-unit measures
4. **Marginal cost**: the extra cost of one more unit — **the most important curve**
5. **MC crosses AVC and ATC at their minimums** (the “exam grade” rule)
6. **LRAC**: the envelope of short-run ATC curves — shows optimal plant size
7. **Economies of scale** explain why large hotel chains exist; **diseconomies** explain why they don't grow forever

Connection: Costs are the **foundation** of supply. Next lecture we use MC to find the **profit-maximizing output**.

Next (Lecture 11, March 20): Companies — Profit Maximization 💰

Exercises

Practice Time! 

Cost curves, short run, and long run.

Exercise 1: Multiple Choice

Question: A tour operator in Lisbon has monthly fixed costs of €10,000 (office lease, software) and variable costs that depend on the number of tours sold. Currently, they sell 200 tours per month with an ATC of €80 per tour. If they increase to 250 tours and ATC falls to €72 per tour, what can we conclude?

- A. Marginal cost of the additional tours is below €72
- B. Marginal cost of the additional tours is above €80
- C. The firm is experiencing diseconomies of scale
- D. Fixed costs have increased

Answer: A

If ATC is **falling** (from €80 to €72), it must be that MC is **pulling the average down**. From the “exam grade” analogy: if your average falls, the new grade must be **below** the current average. So $MC < ATC = €72$ for those additional 50 tours.

Exercise 2: Multiple Choice

Question: In the short run, a small hotel in Sintra has fixed costs of €6,000/month. When occupancy rises from 80 to 90 guests/month, total costs rise from €14,000 to €15,500. What is the marginal cost per additional guest?

- A. €150
- B. €155
- C. €172
- D. €175

Answer: A

$$MC = \frac{\Delta TC}{\Delta Q} = \frac{€15,500 - €14,000}{90 - 80} = \frac{€1,500}{10} = €150 \text{ per guest.}$$

Note: FC (€6,000) is **irrelevant** to MC — it doesn't change! Only variable costs drive the change in TC.

Exercise 3: Open Question

A small surf school in Ericeira has monthly fixed costs of €2,000 (equipment rental, insurance) and the following cost structure:

Students/month (Q)	Variable Cost (VC)
0	€0
10	€800
20	€1,400
30	€1,900
40	€2,600
50	€3,500
60	€4,800

- Calculate TC, AFC, AVC, ATC, and MC for each output level.
- At what output level is AVC minimized? At what output is ATC minimized?
- Verify that $MC = AVC$ at the minimum of AVC, and $MC = ATC$ at the minimum of ATC.
- The school charges €90 per student. At what output level would you expect MC to equal this price? What does this tell us about the profit-maximizing output? (We'll formalize this in the next lecture!)
- In the long run, the school considers expanding to a second location. Explain, using the concept of economies of scale, under what conditions this expansion would lower the cost per student.

Exercise 3: Solution — Part a

Q	FC	VC	TC	AFC	AVC	ATC	MC (per 10 students)
0	2,000	0	2,000	—	—	—	—
10	2,000	800	2,800	200.00	80.00	280.00	80.00
20	2,000	1,400	3,400	100.00	70.00	170.00	60.00
30	2,000	1,900	3,900	66.67	63.33	130.00	50.00
40	2,000	2,600	4,600	50.00	65.00	115.00	70.00
50	2,000	3,500	5,500	40.00	70.00	110.00	90.00
60	2,000	4,800	6,800	33.33	80.00	113.33	130.00

MC calculation example: From Q = 20 to Q = 30: $MC = \frac{3,900 - 3,400}{30 - 20} = \frac{500}{10} = \text{€}50$ per student.

Exercise 3: Solution — Parts b, c & d

b) AVC is minimized at $Q = 30$ (AVC = €63.33). ATC is minimized at $Q = 50$ (ATC = €110.00).

👉 AVC minimum comes **before** ATC minimum — this is always the case because AFC is still pulling ATC down even after AVC starts rising.

c) At $Q = 30$ (AVC minimum): MC just shifted from €60 (at $Q=20$) to €50 (at $Q=30$), and then rises to €70 (at $Q=40$). MC passes through the AVC minimum range, consistent with theory. At $Q = 50$ (ATC minimum): MC = €90, while ATC = €110. Between $Q=40$ and $Q=50$, MC (€90) is still below ATC (€110), pulling it down. At $Q=60$, MC = €130 > ATC = €113.33, so ATC starts rising. The crossing happens between $Q=50$ and $Q=60$. ✓

d) Price = €90. MC = €90 at $Q = 50$. This suggests the profit-maximizing output is around **50 students**. At this level, each additional student brings in exactly as much revenue (€90) as they cost to add. Producing more ($Q=60$) would mean MC = €130 > €90 = price → **losing money on each extra student**.

Exercise 3: Solution — Part e

e) Expanding to a second location would lower the cost per student if the school can achieve **economies of scale**:

- 🤝 **Shared equipment purchases**: buying boards and wetsuits in bulk at a discount
- 💻 **Single booking/marketing system** serving both locations
- 🎓 **Instructor training**: standardized program developed once, used at both sites
- 📈 **Brand recognition**: marketing costs spread over more students

However, expansion could also create **diseconomies** if:

- Coordination between locations is difficult (travel time, communication)
- Quality control drops when the owner can't supervise both sites
- Each beach has unique conditions requiring different approaches

👉 Expansion makes sense if the LRAC **falls** at the combined output — that is, if the firm is still in the **economies of scale** region.

Next Lecture

March 20, 2026: Companies — Profit Maximization 💰

How do firms use cost curves to choose the **optimal output**?

👉 **Spoiler:** The answer involves setting **Price = MC** in perfect competition!

Thank You!

Questions? 🙋

✉ paulo.fagandini@ext.universidadeeuropeia.pt

Next class: Friday, March 20, 2026